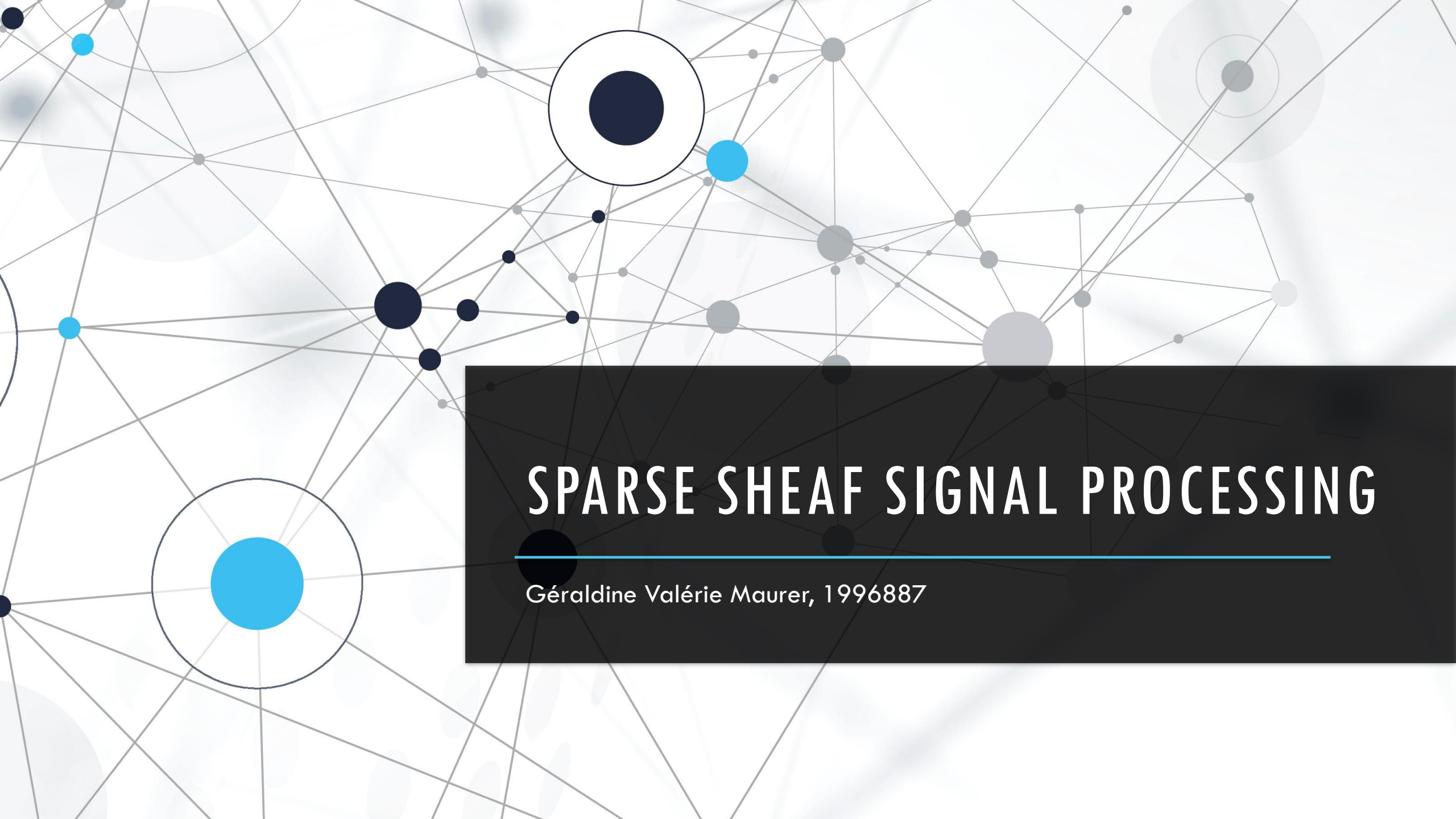


SPARSE SHEAF SIGNAL PROCESSING

Géraldine Valérie Maurer, 1996887

INTRODUCTION

This project explores sparse sheaf signal processing over tangent vector bundles on point clouds in $\mathbb{R}^3$ via wavelet dictionaries.

Structure

1. Implementation of a Vector Diffusion Mapping class
2. Implementation of a Wavelet class
3. Signal compression and signal denoising experiments

Datasets

1. Random point cloud in a $[-0.5, 0.5]^3$ cube in $\mathbb{R}^3$ (300 nodes)
2. Fibonacci sphere in $\mathbb{R}^3$ (300 nodes)
3. Swiss wind speed and directions dataset (151 nodes)
4. ERA5 dataset of wind measurements around the Arctic Circle (100 nodes)

Goal

Comparing the performance on signal compression/denoising tasks using the wavelet dictionaries built from the **connection Laplacian** (VDM) and the **trivial graph Laplacian.**

VECTOR DIFFUSION MAPPING

1. **Graph Construction**: using fixed-radius near neighbours (radius = $\sqrt{\epsilon}$), a weighted graph with edge weights $w_{ij} = G\left(\frac{\|x_i - x_j\|_{\mathbb{R}^p}}{\sqrt{\epsilon}}\right)$ is created, with $G(u) = \exp(-u^2)_{\chi_{[0,1]}}$

2. **Local PCA**: the basis of the tangent plane $T_{x_i}\mathcal{M}$ to the manifold in x_i is approximated using local PCA. For every i, a neighbourhood of nodes $N_{x_i,\epsilon_{PCA}} = \{x_i : 0 < \|x_i - x_j\|_{\mathbb{R}^p} < \sqrt{\epsilon_{PCA}}\}$ is created, with $N_i = |N_{x_i,\epsilon_{PCA}}|$ neighbours of x_i, such that $N_i \geq d$ and $N_i \ll n$. Next, the matrix of shifted neighbour matrices $X_i = \left[x_{i_1} - x_i | \cdots | x_{i_{N_i}} - x_i\right] \in \mathbb{R}^{p \times N_i}$ and the scaling matrices $D_i \in \mathbb{R}^{N_i \times N_i}$ are defined, with

$$D_i(i,j) = \sqrt{K\left(\frac{\|x_i - x_{i_j}\|_{\mathbb{R}^p}}{\sqrt{\epsilon_{PCA}}}\right)} \qquad j = 1,2,\ldots,N_i$$

The matrix $B_i \in \mathbb{R}^{p \times N_i}$ is defined as $B_i = X_i D_i$, with singular values $\sigma_{i,1} \geq \cdots \geq \sigma_{i,N_i}$. The manifold dimension is estimated as follows: for every $i = 1,\ldots,n$, the value d_i is computed as the smallest integer such that

$$\frac{\sum_{j=1}^{d_i} \sigma_{i,j}^2}{\sum_{j=1}^{N_i} \sigma_{i,j}^2} > \gamma$$

The median of these values d_i is the estimated manifold dimension.

VECTOR DIFFUSION MAPPING

1. Graph Construction
2. Local PCA
3. **Orthonormal Basis Approximation**: Let $B_i = U_i \Sigma_i V_i^T$ be the SVD of the previously computed B_i, with $U_i = \left[u_{i_1} | \cdots | u_{i_{N_i}}\right] \in \mathbb{R}^{p \times N_i}$ the orthonormal matrix of the left singular vectors. The orthonormal basis of the tangent vector space $T_{x_i}\mathcal{M}$ is approximated using the first d left singular vectors of U_i:

$$O_i = \left[u_{i_1}, \cdots, u_{i_d}\right] \in \mathbb{R}^{p \times d}$$

4. **Alignment**: the alignment matrices between nodes i and j are computed by first calculating the SVD of $O_i^T O_j = U_{ij} \Sigma_{ij} V_{ij}^T$, and then taking $O_{ij} = U_{ij} V_{ij}^T$
5. **VDM and the Connection Laplacian**: the matrix S is constructed as $S(i,j) = w_{ij} O_{ij}$ for $(i,j) \in E$ and $0_{d \times d}$ otherwise. It consists of $n \times n$ blocks of side $d \times d$. A diagonal matrix of the same size is defined in this way: $D(i,i) = \left(\sum_{j:(i,j) \in E} w_{ij}\right) I_{d \times d}$. The connection Laplacian over the manifold $\mathcal{M}$ is approximated as $L = S - D$, and the normalized connection Laplacian with $L = D^{-1} S - I$.

WAVELETS

1. Compute the **eigen-decomposition** of the Laplacian: $\mathcal{L}x_\ell = \lambda_\ell x_\ell, \quad \ell = 0, \ldots, N-1$
2. Define a **parametric kernel function**: $g(x) = x^h \, e^{-t\,x^p}$ with $t, h > 0$ and such that $g(0) = 0$, $\lim_{x\to\infty} g(x) = 0$ such that it can serve as a band-pass filter. The parameters chosen were $t, p, h = 1$ for the **default** kernel experiments. For the experiments with **adjusted** kernel, the t parameter was heuristically computed as follows: $t = 3/(\max_\ell|\lambda_\ell| + p)$ to adapt to the Laplacian's eigenvalues.
3. Define the sheaf **wavelet operator**: $\psi_{s,a} := T_g{}^s \delta_a$ with $T_g{}^s = g(s\mathcal{L})$ scaled wavelet operator and δ_a the localisation impulse. Applying the operator to a single vertex v gives us:

$$\psi_{s,a}(v) = \sum_{\ell=0}^{N-1} g(s\lambda_\ell)\, x_\ell^*(a)\, x_\ell(v)$$

The **wavelet coefficients** of a signal $f \in \mathbb{R}^N$ are defined as follows:

$$W_f(s,a) = \langle \psi_{s,a}, f \rangle = T_g{}^s f\,(a) = \sum_{\ell=0}^{N-1} g(s\lambda_\ell)\, \hat{f}(\ell) x_\ell(v)$$

WAVELETS

4. Build a **dictionary** of graph and sheaf wavelets. For a fixed scale s, define the diagonal spectral response matrix

$$G(s) = \mathrm{diag}(g(s\lambda_0),\ g(s\lambda_1),\ \ldots,\ g(s\lambda_{N-1})) \in \mathbb{R}^{N\times N}$$

Then the operator $T_g^s = g(s\mathcal{L})$ applied in the frequency domain is:

$$\Psi_s := XG(s)X^\top \in \mathbb{R}^{N\times N}, \qquad X = [x_0 \mid x_1 \mid \cdots \mid x_{N-1}]$$

where X is the matrix of Laplacian eigenvectors (columns), and $X^\top$ is the Fourier transform. The element (v, a) of this dictionary returns the wavelet formula:

$$[\Psi_s]_{v,a} = \sum_{\ell=0}^{N-1} g\,(s\lambda_\ell)\, x_\ell(a)\, x_\ell(v) = \psi_{s,a}(v)$$

For a set $\{s_1, \ldots, s_S\}$ of scales, we get the dictionary:

$$D = [\ \Psi_{s_1} \mid \Psi_{s_2} \mid \cdots \mid \Psi_{s_S}\ ] \in \mathbb{R}^{N\times SN}$$

In the project, I used

5. Use **OMP** to solve the LASSO, basis pursuit problem and create sparse signal representations:

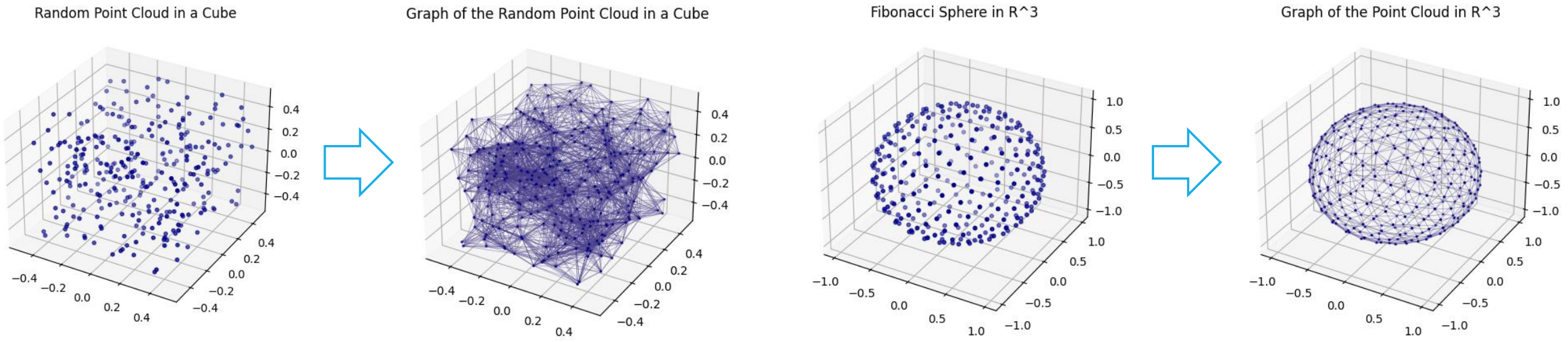
$$\min_x \frac{1}{2}\|\Psi_S x - f\|_2^2 + \lambda\|x\|_1$$

where f is the signal whose sparse representation we are looking for.

SYNTHETIC DATASETS

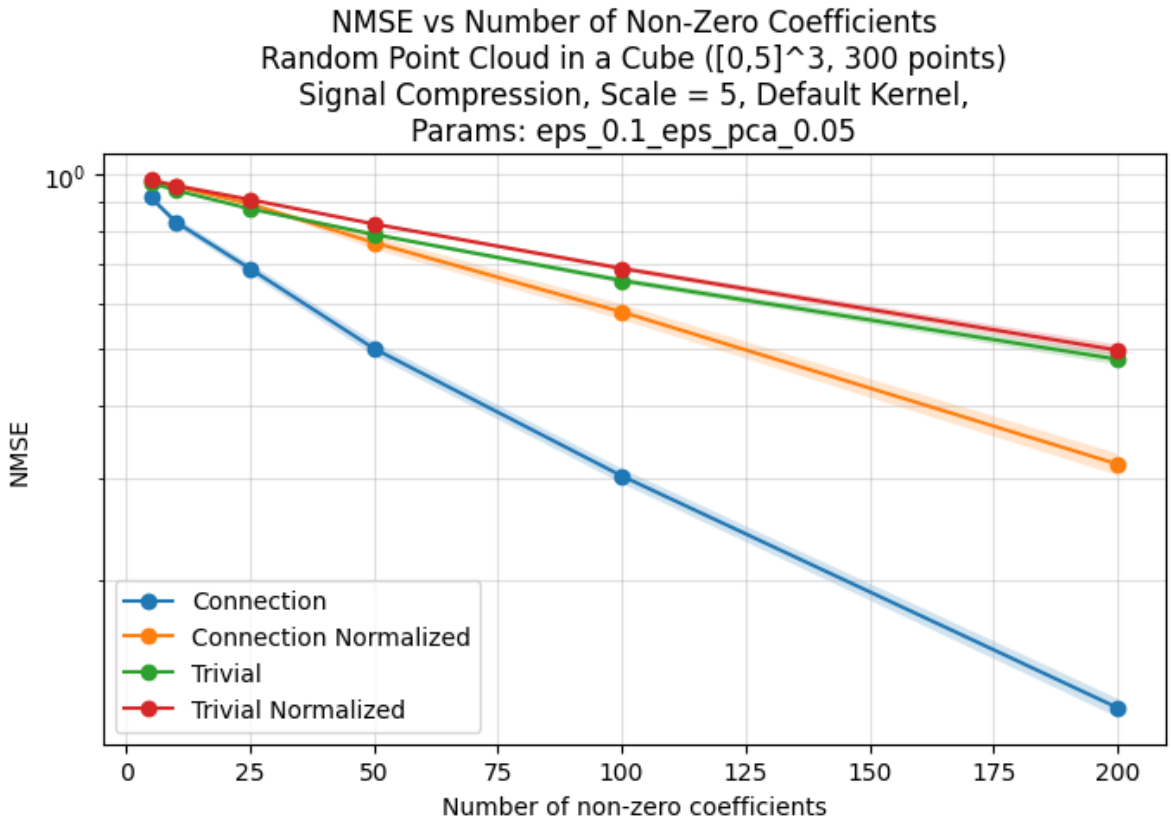
Dataset: a random point cloud in a $[-0.5,0.5]^3$ cube (300 points) and the Fibonacci sphere (300 points)

Signals: samples of a discretized tangent-bundle Gaussian random field: an ambient, spatially-correlated 3D Gaussian vector field (created via *gstools*, with an additional imposed cross-channel covariance Σ) is sampled at the N vertices and projected onto each vertex's 2D tangent plane, giving one $\mathbb{R}^{2N}$ signal per realization, repeated M times to form the sample ensemble.

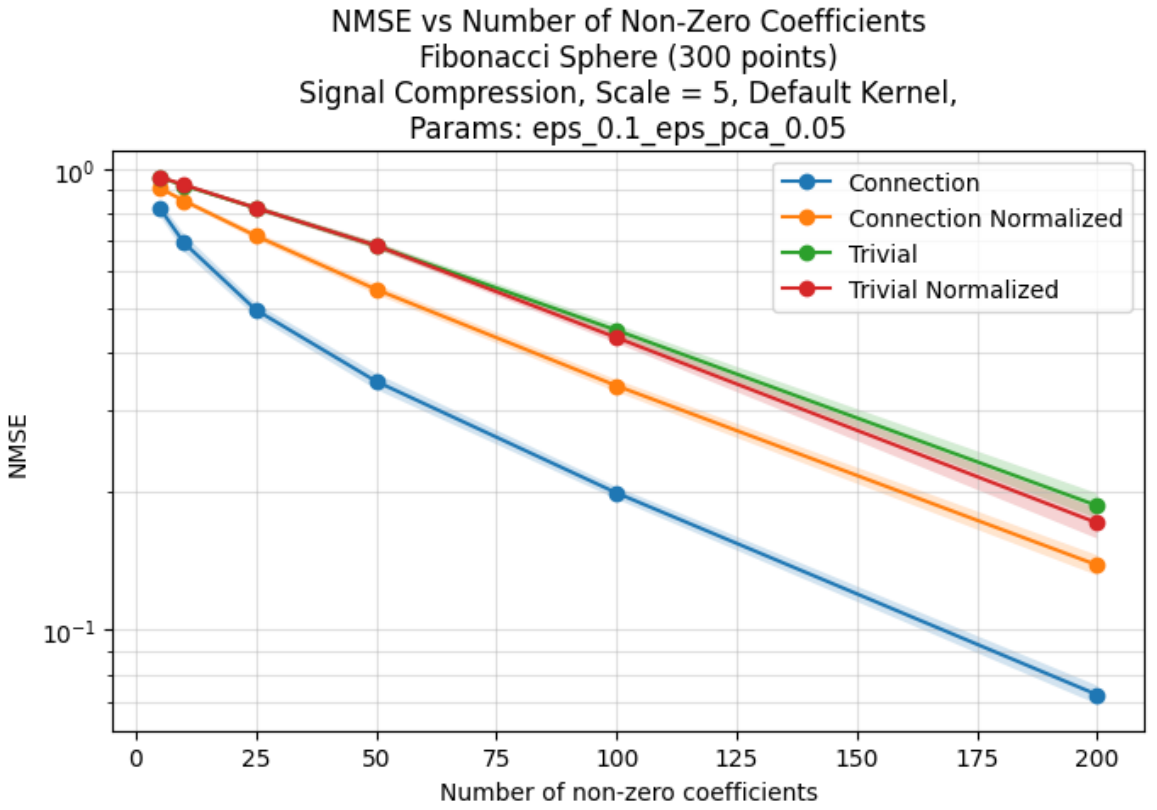


RESULTS: SIGNAL COMPRESSION

Random point cloud in a cube, 20 signals

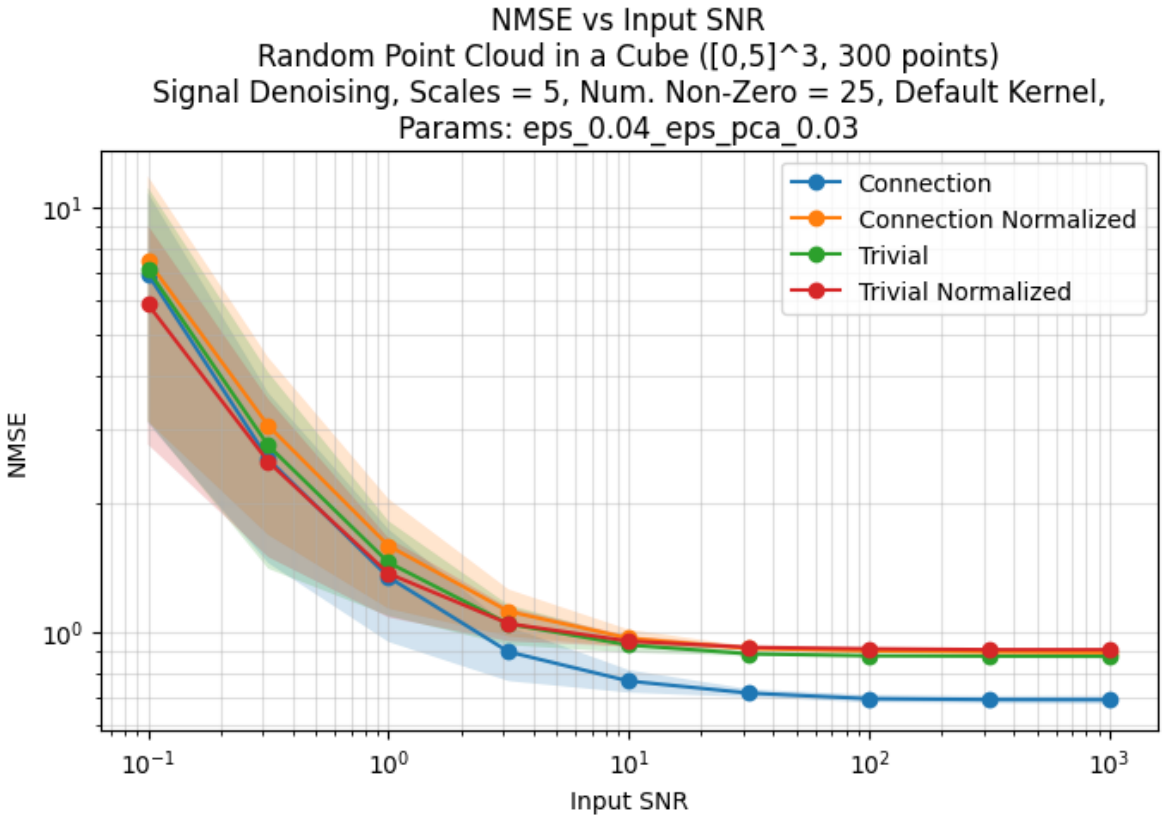


Fibonacci sphere, 20 signals

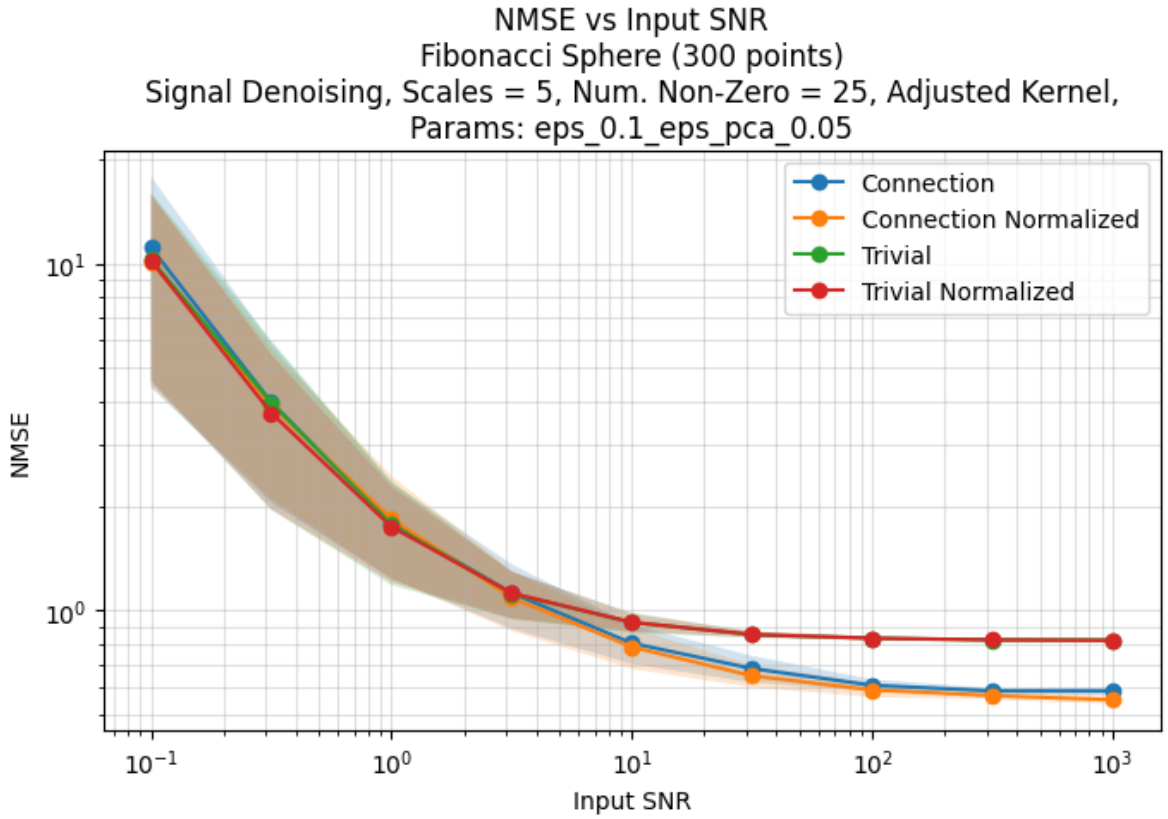


RESULTS: SIGNAL DENOISING

Random point cloud in a cube, 20 signals



Fibonacci sphere, 20 signals

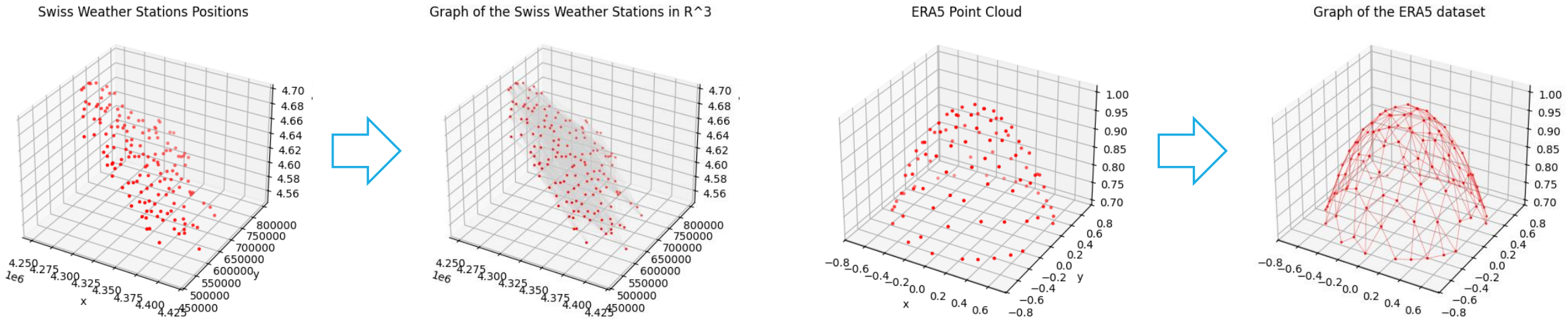


WIND SPEED AND ERA5 DATASETS

Swiss Wind Dataset: the data is downloaded from peakweather.dataset and contains 302 weather station coordinates, filtered such that there are no stations with at least a 0 in wind speed or direction. The final point cloud contains 151 nodes.

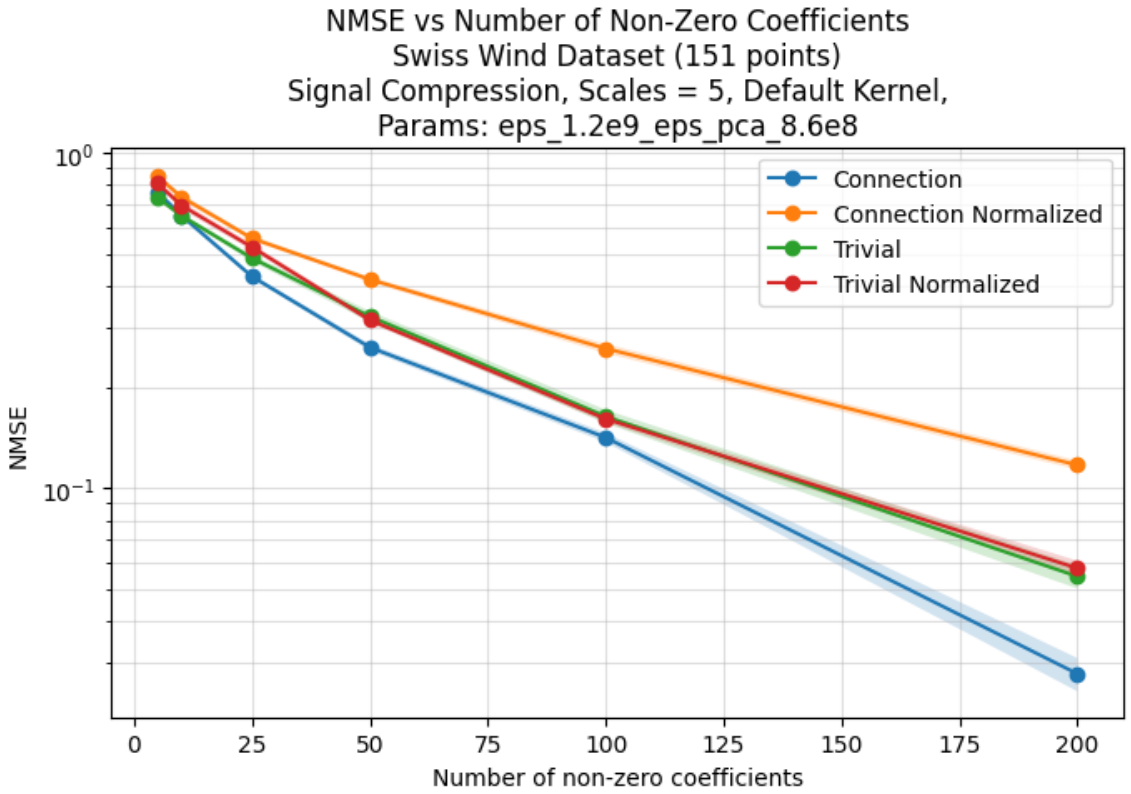
ERA5 Dataset: wind measurements from

Signals: (wind_u, wind_v) values for every station, 20 signals studied for the Swiss wind dataset, 1 from the ERA5 data

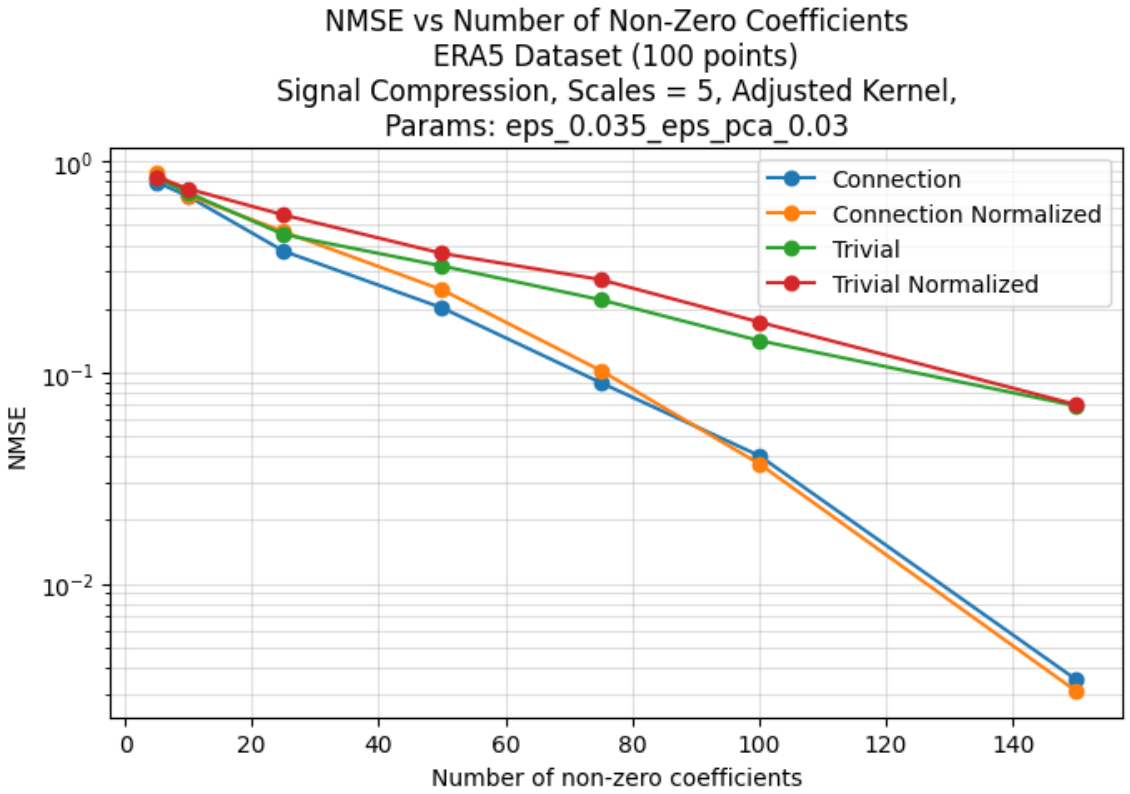


RESULTS: SIGNAL COMPRESSION

Swiss wind dataset, 20 signals

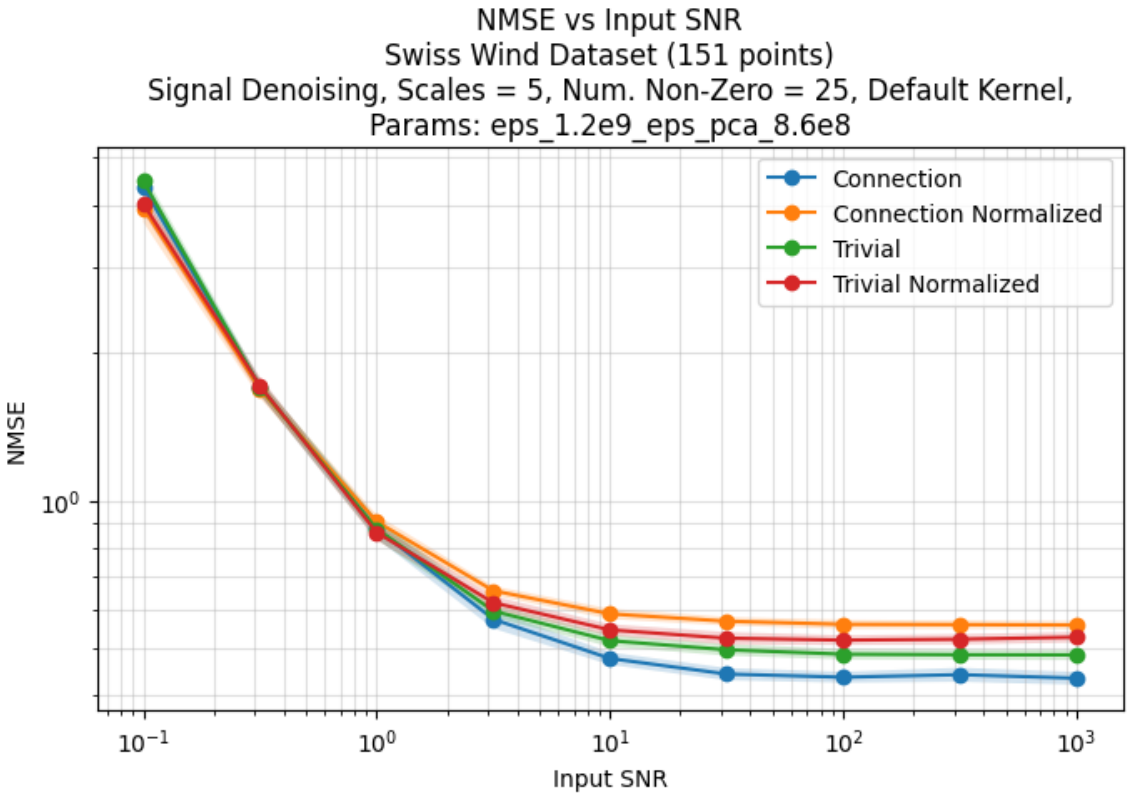


ERA5 dataset, 1 signal

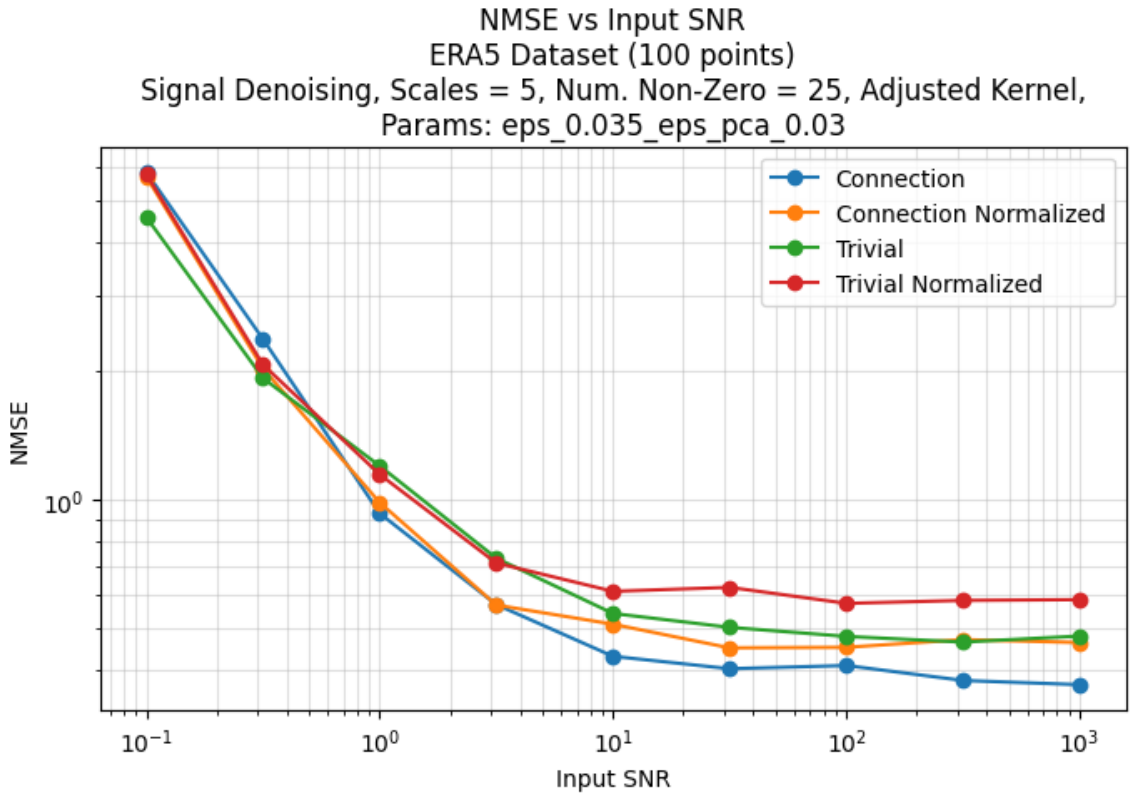


RESULTS: SIGNAL DENOISING

Swiss wind dataset, 20 signals



ERA5 dataset, 1 signal



Thank you for your attention!